

Muhammad Afsah Mumtaz

AI Engineer | Quantitative & Fintech Systems | Multi-Agent Trading

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LinkedIn: linkedin.com/in/muhammad-afsah-mumtaz | GitHub: github.com/NAME0x0 | Portfolio: NAME0x0.github.io

Nationality: Pakistani | Visa: UAE Golden Visa (no sponsorship required) | Open to Dubai / Abu Dhabi (on-site / hybrid) & remote worldwide | Notice period: Immediate

PROFESSIONAL SUMMARY

AI engineer building **quantitative, agentic trading systems** (BSc Hons Information Technology student, Middlesex University Dubai). Designed and shipped **Pantheon-Trades** — a live 11-agent large language model council that debates every prediction-market trade on Polymarket with **Half-Kelly position sizing, self-calibrating agent weights (Platt / isotonic regression), an on-chain audit trail, and a 200-market backtest (Brier 0.149)**. Also fine-tuned a 2B LLM to **82% ARC-Challenge** on a single 4 GB GPU. Strong in Python, FastAPI, Solidity, PyTorch, and risk-aware system design. Seeking an **AI / quantitative / fintech engineering internship or junior role**. UAE Golden Visa holder — no sponsorship required, immediate start.

TECHNICAL SKILLS

Quantitative & Trading: Algorithmic Trading, Backtesting, Strategy Evaluation, Position Sizing (Kelly Criterion), Risk Management, Probability Calibration (Platt / Isotonic), Brier Score, Prediction Markets, Portfolio / Risk Constraints, Avoiding Look-Ahead Bias

AI / Machine Learning: Large Language Models (LLMs), Multi-Agent Systems / Agentic AI, LLM Orchestration, Retrieval-Augmented Generation (RAG), Fine-Tuning (LoRA / QLoRA), Model Evaluation, Prompt Engineering, PyTorch, Hugging Face Transformers, Ollama

Blockchain & Web3: Solidity, Foundry, Smart Contracts, On-Chain Settlement, Arc Testnet, Polymarket CLOB

Languages: Python, Rust, TypeScript, JavaScript, C++, SQL

Backend & Data: FastAPI, Node.js, Express, REST APIs, PostgreSQL, Redis, MongoDB, SQLite

Infra & Tooling: Docker, Git, GitHub Actions, CI/CD, Turborepo / pnpm, Testing (pytest, Foundry)

PROJECTS

Pantheon-Trades — Multi-Agent Quant Trading Platform [Live ↗ pantheon-trades.vercel.app](#)

Python · FastAPI · Solidity / Foundry · PostgreSQL · Redis · Next.js 14 · Multi-LLM (Claude, Gemini, OpenAI, +8)

- Designed and shipped a live platform where an 11-agent LLM council debates every binary prediction-market trade on Polymarket across four structured rounds before execution.
- Implemented risk-aware execution: **Half-Kelly position sizing**, constitutional position caps (5% NAV max), unilateral risk vetoes, and an on-chain proof-of-restraint audit trail recording rejected trades.
- Built self-calibrating agent weights from Brier scores with council probabilities recalibrated via Platt / isotonic regression; validated on a 200-market backtest (**council Brier 0.149**).
- Hardened with 714 Python and 65 Solidity tests across a pnpm / Turborepo monorepo, demonstrating production-grade reliability and testing discipline.

AVA — LLM Fine-Tuning, Evaluation & Inference Lab [github.com/NAME0x0/AVA](#)

Python · PyTorch · Hugging Face · PEFT / QLoRA · RAG · Ollama · GGUF

- Fine-tuned Qwen3.5-2B with a 42 MB QLoRA adapter to **82.0% ARC-Challenge and 92.0% ARC-Easy**, outperforming Llama 3.2 3B-Instruct (78.6%) — all on a single 4 GB GPU.
- Built a reproducible evaluation harness across 17 benchmarks (16,872 tasks) with statistically careful scoring; training peaked at 1.81 GB VRAM in 100 minutes.

MALD — Local AI Knowledge System with Cited RAG

Rust · SQLite FTS5 · HNSW vector indexing · RAG · Ollama

- Built a privacy-first knowledge system as a single Rust binary with hybrid full-text and semantic (HNSW) search and cited RAG chat referencing exact files and lines.

pane — Windows-Native Linux Environment Platform [github.com/NAME0x0](#)

Rust · WSL2 · XRDp · PowerShell · Windows packaging automation

- Developed a Windows-native control surface to create, launch, repair, and manage Linux desktop environments — shipped an Arch-first MVP packaged as pane.exe with automated release packaging.

| WORK EXPERIENCE

Open-Source Contributor — HAGI (Hypercomplex AGI Research) 2026 - Present
github.com/ShmidtS/HAGI | Remote

- Run independent experiments on the experimental branch of an open research architecture exploring Clifford-algebra reasoning in small language models — executing and publishing test results across PyTorch, Rust, and Lean4 and exchanging findings with the contributor team.

Chairman — MDX Computing Society Sep 2025 - Present
Middlesex University Dubai | Dubai, UAE

- Lead the university computing society, directing technical workshops, events, and community initiatives across the IT cohort.

Committee Chair — BCS Student Chapter Sep 2025 - Present
BCS, The Chartered Institute for IT | Dubai, UAE

- Chair the student chapter committee, planning and delivering student-led technical and professional events.

Customer Service Representative Oct 2021 - Jan 2022
transcosmos inc. | Debrecen, Hungary

- Resolved inbound customer issues for a multinational outsourcing operation, consistently meeting response-time and quality targets in a high-volume environment.

| EDUCATION

BSc (Hons) Information Technology Sep 2024 - Jun 2027 (Expected)
Middlesex University Dubai | Dubai, UAE

- Grade 1 (highest grade, UK 20-point scale) in 4 of 6 completed modules; all 6 graded within the first-class band.
- Relevant coursework: Data Science and Machine Learning, Information in Organisations (Databases), Compliance and Professional Project Management, Web Applications and Databases, Programming for Data Communication and Networks.

A Levels Completed
Cambridge International School, Dubai | Dubai, UAE

| AWARDS & COMPETITIONS

- 2nd Place — Odoo Hackathon (team event), Dubai — Nov 2025
- 2nd Place — Curtin University Dubai Hackathon (team event) — Nov 2025

| CERTIFICATIONS

- IBM AI Engineering Professional Certificate — IBM (Aug 2025)
- Blockchain Basics — The State University of New York (Aug 2025)
- AI Strategy and Governance — University of Pennsylvania (Sep 2025)
- Google IT Automation with Python Professional Certificate — Google (Jan 2025)

| ADDITIONAL INFORMATION

Languages: English (Native), Urdu (Native), Arabic (Professional Working), Hungarian (Limited Working)

Work Authorization: UAE Golden Visa holder — no sponsorship required

Availability: Internship, junior, contract, freelance, and full-time. On-site / hybrid in Dubai / Abu Dhabi, UAE, and remote worldwide. Immediate start.